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SoundShaping: Combining Music-Making and Painting in Augmented Reality for Adults with Neurodevelopmental Conditions

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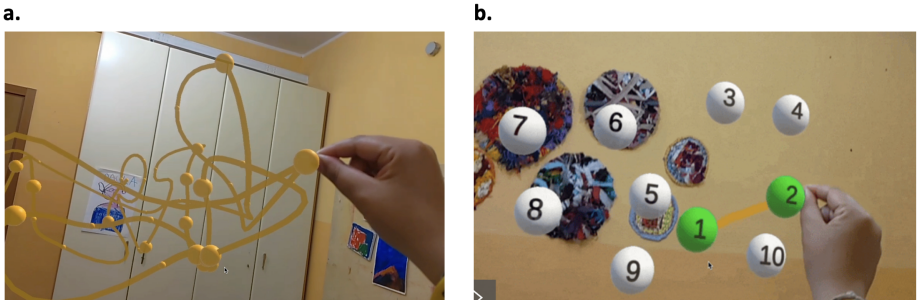

Fig. 1. Main design features of *SoundShaping* **(a.)** Open-ended play mode (view of the user wearing the AR headset) and **(b.)** Structured play mode, derived from the Trail-Making Test (view of the user wearing the AR headset)

Art therapy approaches, such as music-making and painting, can promote attention and emotional regulation abilities while fostering engagement in individuals with Neurodevelopmental Conditions (NDCs). Studies combining music-making and painting activities in immersive, interactive multisensory environments suggest promising benefits to complement traditional approaches. Still, the limited portability of interactive multisensory environments restricts their use in everyday settings, such as care centers. This paper introduces SoundShaping, a head-mounted augmented reality activity that integrates drawing and music-making and is devoted to people with NDCs, particularly adults with moderate to severe intellectual disabilities. The article focuses on the user-centered, iterative design process of SoundShaping, which was co-designed in collaboration with a psychologist specialized in NDCs.

CCS Concepts: • **Human-centered computing** → *Mixed / augmented reality*; *User centered design*; • **Social and professional topics** → *People with disabilities*; • **Applied computing** → *Media arts.*

Additional Key Words and Phrases: Augmented Reality, Art Therapy, Music, Paint, Neurodevelopmental conditions, Intellectual disability

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1 Introduction

Neurodevelopmental Conditions (NDCs), such as autism or Intellectual Disability (ID), involve challenges with emotional regulation, attention, and engagement [1]. Some people display mild learning disabilities[1] and low support needs (e.g., difficulty initiating conversation). Others present with significant learning disabilities, often in combination with complex conditions — for example, minimal language abilities sometimes associated with intellectual disability (ID). The latter population is little represented in current research [21], despite their prevalence. Regarding autism, for instance, 30% of children are non-verbal or minimally verbal [24] and 33% have an ID [17].

Creative practices can help address the challenges people with NDCs face (e.g., emotional and attentional challenges) and are well-suited to varying support needs [7, 22]. Painting and music-making are often used in care centers with various engaging activities, e.g., individual or group-based, improvised or result-oriented [18, 22]. In particular, improvised practices seem promising for individuals with high-support needs [18]. While music-making or painting seem to offer comparable benefits [7], some subjects may prefer either to paint or to make music, prompting caregivers to tailor activities accordingly [22]. As art materials afford different ways of expression, combining various art forms seems promising to work with individuals with NDCs, even though such effect remains understudied [7].

Digital tools with immersive multisensory capabilities, such as Interactive Multisensory Environments (iMSE), offer opportunities to directly combine music-making and painting, potentially enhancing the effects of each. iMSEs are physical spaces augmented with "smart" components (e.g., smart lights, motion sensors). Prior studies have reported positive outcomes for individuals with NDCs, particularly in well-being and social interaction [10, 12, 14, 19, 20]. Two systems, called *SensoryPaint* and *MusicTraces*, combine music-making and painting [6, 20]. *SensoryPaint* enables single users to paint by throwing a ball at a large screen, receiving audio feedback linked to their gestures. Empirical studies showed benefits to helping children with NDCs balance their attention between stimuli [20]. *MusicTraces* is a two-player experience that allows users to paint on a frontal large screen while making music, using bespoke controllers, not yet tested with end-users [6]. However, limitations in portability and high costs remain significant barriers in these projects.

Augmented Reality (AR) headsets can offer comparable levels of immersion with higher portability and are usually well accepted by people with NDCs [3, 16]. They are promising to support the attention of people with NDCs by guiding them toward specific information [15]. However, AR technology remains under-explored for people with NDCs and high-support needs (e.g., minimally verbal with ID) [5], raising questions about which interaction techniques best suit their sensory and cognitive profiles, and which established design guidelines (e.g., simplifying information) [9] are more appropriate for their specific needs. To that end, "magical interactions", defined by [23] as interactions that do not follow real-world physical laws but rather take advantage of the medium's affordances seem promising. Indeed, they have successfully been used with children with NDCs being minimally verbal and with ID in a virtual reality study focusing on making music with their caregiver through a music looper looking like a train, with positive outcomes over engagement and attention [4].

This paper presents *SoundShaping*, a single-user headset-based AR activity that combines drawing and music-making to promote attention, emotional regulation, and engagement in adults with NDCs who display moderate to high levels of learning disabilities and additional complex conditions. A user-centered design process was conducted with a psychologist specialized in NDCs, drawing on the experience gained in prior projects on IMSEs-Interactive Multisensory Environments (*MusicTraces* and *SensoryPaint*). The paper introduces *SoundShaping* and its technology, and focuses on its design requirements, initial design, and the following (re)design iterations.

[1]This paper adopts autism stakeholders' language preferences [8], e.g., identity first-language (e.g., autistic people), no offending terms (e.g., "disorder".)

2 Requirement analysis: from iMSEs to AR headsets

To elicit the core design requirements of *SoundShaping*, an initial design meeting was conducted involving the design and development team and a clinical psychologist who had worked in a care center specialized for people with NDCs for more than ten years. During the meeting, the researchers presented the psychologist with design possibilities drawing upon past iMSEs projects *MusicTraces* and *SensoryPaint*, and the different elements were analysed and thematically classified according to their relevance for the specific user needs and the technological constraints.

Five main design requirements emerged. First, to accommodate diverse user profiles within the target population, music-making and painting should be equally combined, as in *MusicTraces*, and unlike *SensoryPaint*, requiring adaptation of *MusicTraces*' interaction metaphors. Then, to suit adults with high-support needs, the activity should support improvised practices [18], as in *MusicTraces* and one play mode of *SensoryPaint*. Third, to foster attention and engagement, users should be able to paint all around them, a "magical" interaction that leverages AR's affordances beyond the frontal interactions of *MusicTraces* and *SensoryPaint*. Fourth, a single-user activity was preferred to a multi-user activity, in order to facilitate the study of attention, following the guidelines and promising outcomes of *SensoryPaint*. At last, to address iMSE portability issues, no tangible objects were to be used. Instead, users should "magically" interact (draw and make music) with their hands, leveraging AR headset's hand-tracking.

3 Initial design of SoundShaping

3.1 Environment

SoundShaping is a single-user activity that enables adults with NDCs to make music while drawing in their physical space using an AR headset (HoloLens 2) (see Figure 1.a). The activity is open-ended to suit adults with different complex needs. The user can play standing or seated, and experience simplified audiovisual stimuli to accommodate a wide range of sensory and cognitive abilities. The stimuli consist of soothing sounds (marimba) advised in prior projects [4, 11] and minimalistic visuals [9]. The system supports the harmonious creation of music and visuals through rules explained below. The caregiver monitors and controls all the features using a tablet, e.g., the line colors or music evolution (see Section 3.4).

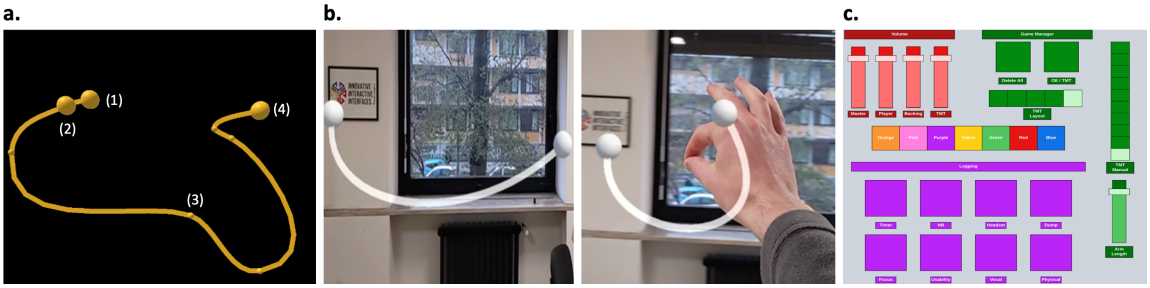

Fig. 2. Design of drawn lines and the tablet interface. **(a.)** Bezier curves used, including music nodes at the start (1), end (4), and along the line (2), as well as line joints (3). **(b.)** Example of a line before (left) and after (right) sculpting. **(c.)** Controls of the tablet interface intended for the caregiver

3.2 The SoundShaping Syntax

Each visual element has a corresponding musical counterpart, and vice versa. Following [2], and *MusicTraces*' study, the experience is structured around three layers: nodes, structures, and creation rules. Nodes are single notes of eight-note chords. Structures are visual combinations of nodes. Rules define instructions and constraints to ensure harmonious creations.

Three design metaphors are used, following *MusicTraces*'. Music nodes are represented by visual spheres. Structures consist of the lines, i.e., minimalistic pipes connecting nodes. Structures can be sculpted upon line completion to organise the sequence of sounds and prevent cluttered visuals. To do so, lines are internally converted into cubic Bézier curves, which users manipulate with control points represented by line joints or the end music nodes (see Figure 2.a).

Three composition rules are used. On the music level, the node height affects their pitch relative to the chord. To support music-making without prior knowledge, the chords are handled by the system. They change every four bars at a tempo of 105 BPM (the pace was adopted heuristically). Eight-note chords were used as a trade-off between the size of the interaction space using the headset and the harmonicity of the music experience. To create a continuously evolving harmonious musical sequence, a variable-length Markov chains model was trained on 37 popular songs to automatically determine the next chord based on previous chords[2]. On the visual level, cubic Bézier curves help ensure smooth transitions between segments, and provide a harmonious visual experience.

3.3 Interactions

The application revolves around three magical interaction modes (summarized in Table 1): creating, sculpting, and playing. Creating consists of drawing lines when users move their hand while making a pinching gesture. Users can create music nodes along the line by rotating their wrist by 30 degrees over a short time interval. This interaction intends to facilitate the intentional creation of music while drawing. To delete a line, users have to pinch two of its nodes, and bring them together. For sculpting, users can manipulate the existing lines and music nodes, either with one or two hands. To do so, they have to pinch the line joints or music nodes, and then move their hand(s) to bend the line accordingly (see Figure 2.b). If users manipulate the music nodes, the corresponding sounds are played. At last, to play, users must hover over the music nodes to play their corresponding sounds.

3.4 Tablet interface

The caregiver can use a tablet interface to monitor the activity or stop it if needed. The interface is structured into four sections (see Figure 2.c). On the top left of the interface, sliders help adjust the sound volume. On the right and top right, the caregiver controls how the experience unfolds (e.g., delete all lines). In the middle, a radio button allows changing the line colors. The bottom of the interface is dedicated to logging. It includes buttons to record the times of significant moments during the experience, for instance related to usability or assistance provided by the caregiver.

3.5 Technology

SoundShaping was developed with Unity engine (version 2021.2.14f1) for the Hololens 2 AR headset. The Mixed Reality Toolkit 3 (MRTK3) package was used inside Unity. The tablet interface was developed using the TouchOSC application. TouchOSC was chosen to externally control the application using the OSC protocol over UDP.

[2]The dataset can be found at this link: https://www.kaggle.com/datasets/taylorflandro/lyrics-and-chords-from-ultimateguitar?select=pop_lyrics_df.csv.

Table 1. Main user interactions

Mode	User action	Visual feedback	Auditory feedback
Creating	Pinch empty space	Temporary node created	Music node played
Creating	Pinch empty space + move hand	Line created with two nodes	Music node played
Creating	Pinch empty space + move and rotate hand	Node added to the line being drawn	Music node played
Sculpting	Pinch line joint + move hand	Line sculpted following user's hand	None
Sculpting	Pinch music node + move hand	Line sculpted following user's hand	Music node played
Sculpting	Hover on line joint	Line joint increases in diameter	None
Playing	Hover on music node	Colour of the music node changes	Music node played

4 Design iterations

We pursued our design process with a psychologist from the care center called Fraternità e Amicizia, specializing in supporting individuals with NDCs. Two remote meetings were held over six weeks, with exchanged emails in between. Discussions focused on improving *ShoundShaping*'s interactions, aesthetics, and functionalities in relation to our research objectives. Before each meeting, videos were shared to illustrate the current state of the application. Textual feedback was analyzed by two of the authors, using a deductive qualitative analysis process related to the activity features and objectives [13].

4.1 First meeting results

The psychologist responded positively to the line-sculpting paradigm, noting its relevance to familiar activities at the care center (e.g., modeling clay). Moreover, he suggested its potential for other conditions, such as upper limb rehabilitation, by encouraging users to perform various movements.

Regarding AR interactions, the pinch gesture was deemed appropriate, though potentially challenging for users with severe motor issues. An alternative gesture (i.e., clenching a fist) was recommended to cater to diverse sensory needs. A wrist rotation gesture was also proposed for making music without drawing. However, some users might struggle with it, considering their cognitive and sensory profiles. Thus, this feature was deemed not essential for the experience and left as an optional action available to enrich the experience of the most proficient users. The possibility of drawing with either one or both hands was also explored. It was agreed that drawing with both hands should be available so that users could draw with one or two hands based on their profiles.

Finally, the psychologist validated the importance of enabling caregivers to tailor application parameters to individual needs to improve user experience and facilitate adoption in care settings.

4.2 Second meeting results

The psychologist confirmed the system's potential to support the attention, emotional regulation, and engagement of adults with NDCs and high-support needs. In particular, interactions seemed promising to support fine motor skills and encourage making novel movements.

To assess attention during the experience, the psychologist proposed adapting the Trail Making Test (TMT) —a standardized pen-and-paper tool already in use at the care center to assess attention — into an AR format (TMT-AR), given its similarities with our activity (i.e., connecting nodes). Administering the TMT-AR before and after the open-ended mode could provide insight into attention levels. At the same time, comparisons with traditional TMT results may help validate the suitability of the TMT-AR test. Additionally, eye-tracking was recommended to obtain quantitative measures of attention during testing.

5 Redesign of SoundShaping: Integration of the Trail Making Test

Following the second design meeting, the TMT-AR play mode was added to assess users' attention (see Figure 1.b). Users connect eight numbered spheres in ascending order by drawing lines. Upon successful connection, the node turns green and plays a corresponding musical sound. To avoid discouragement, errors produce no negative feedback, though users must return to the last correct node before proceeding.

Adapting the pen-and-paper TMT test for our headset-based AR setup required making trade-offs concerning the three-dimensional context (e.g., limited field of view). The node positioning and distance relative to each other were adapted to the field of view of the HoloLens 2 headset based on a pen-and-paper TMT test the caregiver forwarded us. New controls were added to the tablet to give caregivers control over the TMT-AR mode, including a "'TMT Manual" slider that advances the task if users meet difficulties or technical issues (see Figure 2.d). Eye-tracking integration was deferred due to time constraints. It will be considered in future prototypes.

6 Conclusion and Future Work

This paper presented the user-centered design process of *SoundShaping*, a head-mounted AR activity combining music-making and painting that was designed to promote attention, emotional regulation, and engagement in adults with severe forms of neurodevelopmental conditions (NDCs). Design requirements were identified in collaboration with a psychologist specializing in NDCs and built upon the experience and guidelines established in previous studies [6, 20]. An iterative co-design process enabled progressive revisions and enhancements to the user experience (UX) design, resulting in a robust prototype now ready for clinical evaluation.

The research reported in this paper offers three main contributions. To our knowledge, *SoundShaping* is the first system to combine music-making and painting for adults with NDCs using a head-mounted AR setup, extending previous work constrained by the spatial characteristics of smart multisensory environments [6, 20]. Second, novel interaction methods were introduced, including magical AR interactions (e.g., sculpting lines), tailored to profiles of our target population [23]. Third, a standard clinical assessment of attention, the Trail Making Test, was adapted into AR in collaboration with a clinical psychologist, offering an embedded AR evaluation method.

Future work will mainly focus on empirical evaluation, which will be organized in two stages. An initial exploratory study will be performed at a care center, involving four people with NDCs who display mild to moderate ID, and their caregivers. The objective will be to test if the system is truly usable by this special population and is appropriate for a care context. If this testing yields positive outcomes, a wider and prolonged empirical study will be conducted with approximately 15 adults with NDCs recruited in the same context. They will be involved over multiple sessions of *ShoundShaping* use to assess its impact on attention, engagement, and emotional regulation. In this context, the TMT-AR test will be used before and after the open-ended play mode, and compared with the pen-and-paper TMT, to assess attention, validate its use, and further refine it.

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